

Hyperspectral Image Denoising via Dynamically Adjusted Regional Rank Guided by Adaptive Superpixel Segmentation

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Abstract—Hyperspectral image (HSI) is inevitably contaminated by mixed noise during acquisition and imaging, which severely affects subsequent applications. Therefore, HSI denoising is an indispensable part of HSI preprocessing. To solve this problem, this paper proposes an HSI denoising method based on adaptive superpixel segmentation and mathematically grounded regional rank regulation. To avoid the influence of empirical parameters on clustering accuracy, we design an adaptive segmentation framework. This framework autonomously determines the optimal number of homogeneous regions by evaluating internal pixel regional uniformity and boundary discrimination. Instead of relying on a fixed empirical rank, the rank of the CP tensor decomposition for each region is dynamically adjusted. Specifically, the detail complexity of each region is explicitly quantified by combining its spectral variance and spatial gradient energy. The adaptive rank is then mathematically determined by a formulated regression function linking this computed complexity to the region's estimated noise level. Furthermore, a masked completion strategy is introduced to apply CP decomposition to these irregularly shaped superpixels, accurately extracting clean spatial-spectral factors without artifacts. Finally, the Alternating Direction Method of Multipliers (ADMM) optimization framework is used to make the objective function quickly converge. Extensive simulated and real experimental results demonstrate that the proposed method achieves satisfactory denoising performance and robustness. Code will be released at <https://qzhang95.github.io>

Index Terms—Hyperspectral image, mixed noise removal, superpixel segmentation, low-rank tensor decomposition, adaptive.

I. INTRODUCTION

HYPERSPECTRAL image (HSI) is a kind of high dimensional remote sensing data, which contains a large

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amount of band information and has high spectral resolution. It makes HSI more widely used than natural images in lots of applications [1], [2], [3], [4]. However, due to the complex acquisition and imaging environment, the HSI data obtained by the sensor is usually inevitably contaminated by various noises [5], which may reduce the imaging quality of HSI and affect the results of subsequent tasks. Therefore, HSI denoising is an important part of HSI preprocessing.

With the remarkable improvement in computing power of computers, denoising methods can be classified into two categories: model-driven and data-driven. The data-driven methods [6], [7], with the outstanding ability for information learning, can make better use of the spatial-spectral information. Although these methods have reached a lightweight level, on account of the relatively weak generalization ability of data-driven methods, model-driven approaches based on the inherent characteristics of HSI remain the mainstream.

Due to the three-dimensional structure of HSI, it naturally has prior properties such as low rank, smoothness, non-local self-similarity and sparsity [8], [9]. In the early stage, it is common to treat each band of the HSI as a grayscale image, and then recover each band separately [10], [11], which ignores the correlation information between the bands of HSI. In order to solve the above problems, matrix factorization methods expand the 3D HSI data into a 2D matrix along the spectral direction, and separate the low-rank part from the sparse part by using the low-rank property [12], [13]. In order to take advantage of the smoothing characteristics of images, related works have also proposed the combination of low-rank matrix factorization and total variation regularized TV [14]. Wang et al. [15] assumed to be located in a low-rank subspace, and developed the LRTV method. However, the direct expansion of matrix factorization methods ignore the 3D structure information of HSI.

Therefore, low-rank tensor decomposition methods for high-dimensional data have gradually become the mainstream strategy [16], [17]. Among them, Tucker decomposition and CANDECOMP/PARAFAC (CP) decomposition are two commonly methods [18]. Tucker decomposition improves the efficiency of the decomposition by effectively characterizing the coefficient image and the orthogonal basis matrix [19], [20]. He et al. [21] developed a method for mining the low-rank properties of reduced images (NGMeet) using non-local similarity priors. Tucker decomposition needs to separately set

the ranks of 3 modes, and the physical meaning of the core tensor cannot be directly related to the material composition in the noise image [22], [23]. To avoid the above problems, CP tensor decomposition provides another idea. By characterizing the high-dimensional tensor as the sum of several rank-one quantities, CP decomposition can effectively capture the spatial-spectral correlation of HSI. CP decomposition has the advantages that it only needs to set a rank number. It does not need to explicitly design the sparsity of the core tensor, and has a unique solution. For example, Xue et al. [24] constructed a method (NLR-CPTD) to achieve denoising using non-local similarity. However, this simple block method ignores the spatial local correlation of pixels, and cannot well reveal the global spatial similarity.

In recent years, the subregion method based on superpixel segmentation has shown favorable advantages. It can fully mine the spectral-spatial information of HSI and ensure the accuracy of segmentation by superpixel information. Zhang et al. [25] (MSSR) solved the problem of determining the optimal superpixel size. Jiang et al. [26] (Super-PCA) extracted the dominant components of an image. Peng et al. [27] (S3LRR) simultaneously explored the spectral-spatial domains of low-rank properties, and Mei et al. [28] introduced superpixel segmentation (RGBM) into the low-rank representation to preserve the spatial correlation of pixels in homogeneous regions. Han et al. [29] proposed the method of completing irregular homogeneous tensors into regular tensors (ITLRR), which effectively solved the problem that the homogeneous regions segmented by superpixel could not be decomposed by low-rank tensor. For the current superpixel methods, the segmentation number of homogeneous regions needs to be set manually. So that the segmentation result depends heavily on the segmentation number, and then the self-similarity and low-rank characteristics of homogeneous regions are not fully explored. However, a key difficulty in applying tensor decomposition to superpixel-based methods is the shape mismatch: CP and Tucker decompositions require regular 3D arrays as input, while the homogeneous regions obtained from superpixel segmentation are irregularly shaped. Existing works either force regular grids that break object boundaries, or fall back to matrix factorization that discards the multi-linear spectral structure. To address this, we introduce a Masked Tensor Construction strategy that embeds each irregular superpixel into a regular tensor via a binary mask, so that the CP decomposition can still be applied to arbitrarily shaped regions without boundary distortion. In order to solve the above problems, this paper proposes an HSI denoising method based on dynamic adjustment of region rank under the guidance of adaptive superpixels segmentation. The main contributions of the proposed method can be summarized as follows:

- The proposed method adaptively segments HSI into multiple homogeneous regions of superpixel blocks, aiming to mine the global spatial self-similarity.
- The detail complexity of each homogeneous region is calculated from the spatial and spectral dimensions, and the noise level is estimated. Then the rank of the

homogeneous region is dynamically adjusted, to explore the local low-rank property of each region.

- The proposed method uses the dynamically adjusted rank number to perform low-rank tensor CP decomposition for each homogeneous region, and then maps and aggregates the reconstructed regions to obtain the denoised HSI.

II. METHOD

HSI can be viewed as a 3D cube tensor $\mathcal{Y} \in \mathbb{R}^{M \times N \times B}$, where M , N and B are the number of rows, columns and bands of the image, respectively. In the actual acquisition and imaging process, HSI is often contaminated by a variety of mixed noise, such as Gaussian noise, impulse noise, stripe noise and dead line. By extending the classical additive noise model, the degradation process of HSI can be expressed as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} \quad (1)$$

Generally, the HSI denoising process can be regarded as a classical separation model, which separates the clean part $\mathcal{X} \in \mathbb{R}^{M \times N \times B}$ and the mixed noise part $\mathcal{S} \in \mathbb{R}^{M \times N \times B}$ from the observed 3D tensor $\mathcal{Y} \in \mathbb{R}^{M \times N \times B}$. The overall flow of the proposed method is shown in Fig. 1, which can be divided into four steps: Homogeneous regions adaptive segmentation, CP tensor decomposition with dynamic regulation region rank, irregular tensor approximation, and iterative optimization. The specific descriptions of each step are given as follows.

A. Homogeneous Regions Adaptive Segmentation

1) *Entropy Rate Superpixel Segmentation*: As shown in Fig. 1, the proposed method uses principal component analysis (PCA) to extract the first dominant component from the 3D cube image, and reduce the dimension of the 3D tensor into a 2D matrix $Y \in \mathbb{R}^{M \times N \times I}$. PCA is a dimensionality reduction method based on information theory, which can effectively preserve the main spatial-spectral information in [26]. It performs feature enhancement by maintaining original spatial characteristics and intrinsic spectral redundancy, yielding components with the largest variance that reflect the main structures of HSI. Compared to direct 3D segmentation, utilizing these dimension-reduced components significantly reduce the calculation of subsequent tasks while ensuring the accuracy of segmentation. Superpixel mainly represents the spatial characteristics of HSI, and the spectral dimension reduction and region merging through PCA are indirectly completed. The homogeneity division is realized by combining spectral information. Entropy rate superpixel segmentation (ERS) is an image segmentation method [30]. It generates compact, homogeneous and uniform size superpixel by maximizing the entropy rate and balance term of random walk. The key idea is to transform the segmentation problem into a submodular function maximization problem by a greedy algorithm. Map the first principal component image to a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V}$ as the set of vertices and $\mathcal{E}$ as the set of edges. The vertices represent pixels, while the weights indicate the

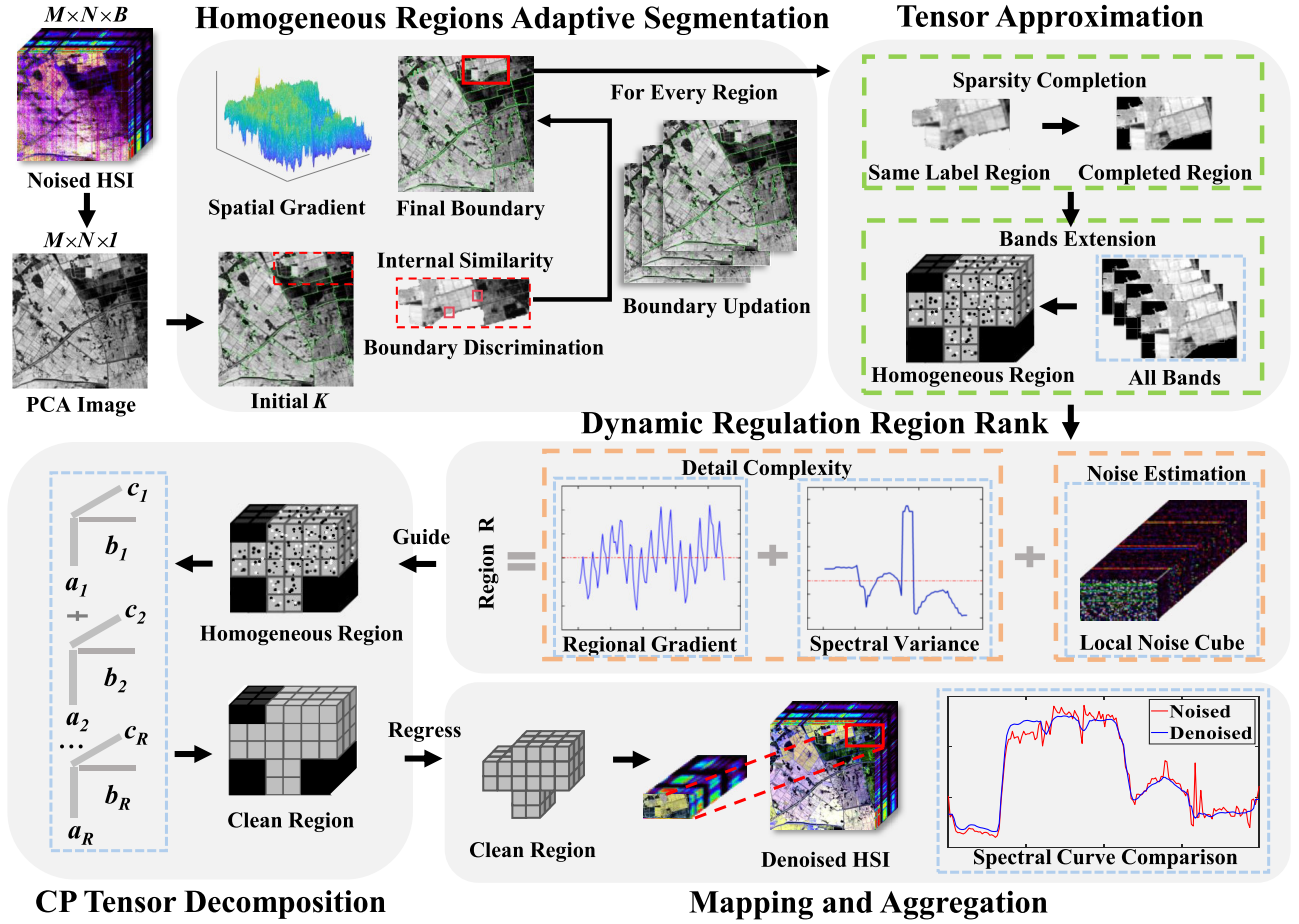


Fig. 1. Flowchart of the proposed method.

similarity between pixels. The random walk model $X = \{x_t\}$ has transition probability:

$$P_{ij} = \Pr(x_{t+1} = v_j | x_t = v_i) = \frac{\mathcal{W}_{i,j}}{\mathcal{W}_i} \quad (2)$$

where $\mathcal{G}$, μ is a stationary distribution. $\mu_i = \frac{\mathcal{W}_i}{\mathcal{W}_T}$, $\mathcal{W}_i = \sum_k \mathcal{W}_{i,k}$, $\mathcal{W}_T = \sum_i \mathcal{W}_i$. $\mathcal{W}_{i,j}$ represents the pixel similarity, the higher the pixel similarity within the superpixel, the more concentrated the P_{ij} distribution and the lower the entropy rate. In order to quantify the residual uncertainty of random variable $X = \{x_t\}$, according to the entropy rate theory of Markov chain, the entropy rate calculation term is introduced as follow:

$$\mathcal{H}(A) = H(x_{t+1} | x_t) = \sum_i \mu_i \sum_j P_{i,j} \log P_{i,j}(A) \quad (3)$$

where $A \subseteq \mathcal{E}$ is a subset of $\mathcal{E}$ and represents the edges that are preserved during the segmentation process. Superpixels are generated by choosing A such that the graph G is divided into K connected subgraphs, that is, K superpixels. In order to avoid the problem of uneven size of the segmented superpixels, a balance term is introduced:

$$\mathcal{B}(A) = H(Z_A) - N_A \quad (4)$$

where Z_A is the size distribution of superpixels and $H(Z_A)$ is the distribution entropy, using to encourage the size uniformity

degree. N_A is the number of connected components, using to penalize small regions. Entropy rate $\mathcal{H}(A)$ and balance term $\mathcal{B}(A)$ are monotonically increasing submodular functions. $\mathcal{H}$ and $\mathcal{B}$ are jointly modeled as a submodular function $\mathcal{F}$, and the greedy algorithm is used to solve Eq (5):

$$\max_{A \subseteq \mathcal{E}} \mathcal{F}(A) = \mathcal{H}(A) + \mathcal{B}(A) \quad (5)$$

2) *Adaptive Determination of the Number of Segmentations:* Most of superpixel segmentation methods usually need to manually set the number of superpixels K , which depends on experience and has great defects. The segmentation results may not really fit the real homogeneous regions, which seriously affects the accuracy of subsequent low-rank representation work. To solve this problem, this paper propose an adaptive segmentation framework based on segmentation quality evaluation system, which adaptively determines segmentation numbers by calculating the quantitative indicators inside and at the boundaries of superpixels. For the first principal component $Y \in \mathbb{R}^{M \times N \times I}$, the original segmentation number K_0 is initialized according to its spatial gradient, and it is divided into homogeneous regions K_0 by ERS. And a label map La is generated to distinguish homogeneous regions. At the current segmentation number, the mean variance of the internal pixel spectral dimension is calculated for each region. The smaller the mean value is, the smaller the fluctuation of the

spectral curve of most pixels is, and the more similar the values of each band are, the more uniform the pixels in the interior of the segmented region are in the spectral dimension, which can effectively mine the spectral consistency of the whole image. The single-pixel variance is denoted by $\text{Var}(\mathcal{SP}_k)$, and the feature vector of pixel p is denoted by $\mathcal{Y}(p)$. Then for each superpixel $\mathcal{SP}_k (k = 1, 2, 3 \dots K)$, its Regional Uniformity (RU) is obtained by the following equation:

$$\text{RU} = \frac{1}{K} \sum_{k=1}^K \text{Var}(\mathcal{SP}_k) \quad (6)$$

The spatial gradient of the current La is $\nabla \text{Spa}(La) = \sqrt{(dxLa)^2 + (dyLa)^2}$, and dx and dy are difference operators in the first and second dimensions. In the spectral dimension, the intensity of the spectral change at the superpixel boundary can be accurately measured by the boundary discrimination. When the spatial gradient $\nabla \text{Spa}(La)$ is not 0, that is, the set $\partial\Omega = \{(i, j) \mid \|(dxLa)_{i,j}\| \&\& \|(dyLa)_{i,j}\| \neq 0\}$ is satisfied, the current position can be considered as the segmentation boundary. In the spectral dimension, the gradient magnitude at the current position is calculated by:

$$\|\nabla \text{Spe}\mathcal{Y}(p)\|_2 = \sqrt{\sum_{b=1}^{B-1} (dz\mathcal{Y}(i, j, b))^2} \quad (7)$$

If the gradient magnitude Eq (7) at the current position is larger, it means that the boundary discrimination is better and the superpixel boundary is more accurately aligned with the true edge in the image. The Boundary Discrimination (BD) of each boundary pixel $p = (i, j) \in \partial\Omega$ is determined as follows:

$$\text{BD} = \frac{1}{\partial\Omega} \sum_{p \in \partial\Omega} \|\nabla \text{Spe}\mathcal{Y}(p)\|_2 \quad (8)$$

Finally, the segmentation quality evaluation (SegQ) function can be constructed by combining the internal pixel regional uniformity (RU) and segmentation boundary discrimination (BD):

$$\text{SegQ} = \alpha \left(1 - \frac{\text{RU}(La)}{\text{Var}_g} \right) + \beta \frac{\text{BD}}{\max(\|\nabla \text{Spe}\mathcal{Y}(p)\|_2)} \quad (9)$$

$$\text{Var}_g = \frac{1}{MN} \sum_p \|\mathcal{Y}(p) - \mu_g\|_2^2 \quad (10)$$

where the global spectral mean μ_g represents the average spectral curve of all pixels over all bands, reflecting the overall spectral distribution of the image. In the segmentation quality evaluation system, it is used to calculate the global variance Var_g as a normalization factor to eliminate the influence of the image scale itself. When the segmentation quality is high enough, the segmentation number K at this time is taken as the final result, and the corresponding label map La is input into the subsequent irregular tensor completion and approximation step as the coordinate index of the homogeneous region.

By constructing a segmentation quality evaluation system, the data waste caused by manually determining the number of segmenting is effectively avoided. The entropy rate maximization ensures the homogeneity within the superpixel, and the generated superpixel boundary accurately fits the natural edge

of the object, which is significantly better than the traditional block clustering method.

B. CP Decomposition With Dynamic Regulation Region Rank

1) *CP Decomposition*: CANDECOMP/PARAFAC (CP) is a powerful low-rank tensor decomposition model that decomposes a tensor into a sum of several Rank-1 tensors by explicitly capturing multilinear latent factors. For example, the tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ can be expressed as:

$$\mathcal{X} \approx \hat{\mathcal{X}} = \sum_{r=1}^R \mathcal{X}_r = \sum_{r=1}^R \sigma_r \cdot a_r \circ b_r \circ c_r \quad (11)$$

where $\hat{\mathcal{X}}$ is the rank R approximation of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ denotes the rank tensor, and $a_r, b_r, c_r \in \mathbb{R}^{I_n}$ is the joint norm vector of the mode n -space of $\mathcal{X}$, $n = 1, 2, 3$. The CP decomposition is used to compute $\hat{\mathcal{X}}$.

The traditional CP decomposition model needs to manually specify the global rank R of the whole image, however, the determination of the tensor rank is a challenging NP-hard problem. In addition, each homogeneous region has good pixel uniformity and spectral consistency after segmentation by our method. If the rank R is set manually based on experience, it may greatly waste the natural advantages of homogeneous regions in denoising tasks. Therefore, the corresponding optimal rank must be selected according to the different information contained in each homogeneous region. In order to solve this problem, this paper proposes a dynamic adjustment strategy for the regional optimal rank R . It can not only effectively infer the underlying multilinear factors with low-rank constraints, but also effectively avoid overfitting through the restriction of rank space, which is described as follows.

2) *Dynamic Adjustment of the Regional Rank R* : For each homogeneous region segmented in the previous step, the optimal rank R of the region is dynamically adjusted according to the detail complexity and noise level of the current region, and then the features are mapped to the rank space. The proposed method exploits the ability of CP decomposition to simultaneously encode joint spatial-spectral information, so that the spatial-spectral strong correlation of HSI is accurately captured. The spectral curves differ significantly from band to band and may have multiple reflection peaks, steep changes, or noise. The regions with large spectral variance usually contain more distinguishable features, and the effective information to be represented is more complex, which means that more rank-one quantities are needed for accurate reconstruction. Regions with small variance may be saturated with information redundancy or dominated by noise, implying that less rank-one piece is required to eliminate redundancy. In order to fully characterize the detail complexity of the current region, it is necessary to combine the spectral variance with the frequency domain gradient operator to comprehensively estimate the detail complexity of the current region.

In order to make the regional optimal rank more consistent with the real value, it is necessary to further estimate the current noise level and complete the robust correction of noise. A large number of experimental results show that the minimum

eigenvalue of the covariance matrix of the weak texture region in the image is highly consistent with the real noise level, because the signal energy of the weak texture block is close to 0. So the image and gradient map are divided into blocks, and the total gradient energy of each block is used to measure the texture intensity of each block [31]. The operator is used to calculate the horizontal and vertical gradient of the image, highlighting the high gradient area. The weak texture area is filtered out by iteration, and the minimum eigenvalue of its covariance matrix is calculated to estimate the noise level [32]. By mining the spectral-spatial joint features, the region rank R_i can be dynamically optimized while balancing the fitting accuracy and computational efficiency:

$$R_i = \text{clip}(\tau(\rho \text{Com}(\mathcal{X}_i) - \text{NoiEst}(\mathcal{X}_i)), R_{\min}, R_{\max}) \quad (12)$$

where $\mathcal{X}_i (i = 1, 2, 3 \dots K)$ represents the region tensor to be decomposed. $[R_{\min}, R_{\max}]$ represents the rank space. Com represents the function to calculate the detail complexity of the current region by Eq (13). NoiEst represents the noise estimation of the current region. τ and ρ are both the model parameters. ω represents the fusion weight resulting from the combination of spectral variance $\text{mean}(\text{Var}(\mathcal{X}_i))$ and frequency domain gradient operators $\text{GradOper}(\mathcal{X}_i)$.

$$\text{Com}(\mathcal{X}_i) = \omega(\text{mean}(\text{Var}(\mathcal{X}_i)) + \text{GradOper}(\mathcal{X}_i)) \quad (13)$$

Since exact CP rank determination is NP-hard, the proposed method does not pursue a theoretically optimal rank. Instead, the bounded regression in Eq (12) confines the estimated rank within a predefined low-rank interval $[R_{\min}, R_{\max}]$, which prevents the decomposition from diverging and keeps the solution stable. Although this strategy yields a sub-optimal rather than optimal rank, consistent performance across six satellite sensors with different spectral resolutions and noise conditions indicates that it is practically effective and not limited to a particular dataset.

C. Irregular Tensor Approximation and Completion

Using the result of superpixel segmentation as an index, the original data can be divided into multiple irregular homogeneous regions, namely $\mathcal{Y}^O = \mathcal{Y}_1^O + \mathcal{Y}_2^O + \dots + \mathcal{Y}_i^O + \dots + \mathcal{Y}_K^O$. Where K is the segmentation number, denotes the i_{th} homogeneous region, and the symbol $\oplus$ denotes the aggregation of multiple irregular homogeneous regions into the whole HSI.

At present, the mainstream low-rank decomposition methods generally require that the input tensors have regular shapes. Specifically, the optimization objective of CP decomposition is usually to minimize the norm error between the reconstructed tensor and the original tensor, which requires the calculation of all tensor elements. If the original tensor has missing values due to irregularity, the mode-1 expansion of the tensor is incomplete, and the pseudo-inverse cannot be calculated properly, the global error cannot be calculated. Therefore, it is necessary to realize the low-rank approximate completion of irregular 3D data by completion and mask mechanism. To this end, this section introduces a completed irregular tensor $\mathcal{C}_i \in \mathbb{R}^{h_i \times w_i \times B_i}$ to fill the irregular part. The original position of each superpixel block is recorded by

coordinate index, and the completed regular tensor is generated by concatenating the original block and the completed block:

$$\mathcal{Y}_i = \mathcal{Y}_i^O \oplus \mathcal{C}_i \quad (14)$$

Low-rank approximation of the completed regular tensor can extract a low-rank representation corresponding to the original irregular tensor. The completion part $\mathcal{C}_i$, which is only used to construct regular tensors to satisfy the reconstruction requirements of low-rank decomposition, is initialized to zero and no optimization constraints are imposed. $\mathcal{C}_i$ does not participate in the actual computation and can ensure that the global low-rank property is only driven by the original region $\mathcal{Y}_i^O$. Note that this is different from simple zero-padding. Because $\mathcal{M}_i$ is incorporated into the fidelity term of Eq (15), the reconstruction error is computed only at the positions where $\mathcal{M}_i = 1$. During ALS updates, the gradient with respect to the factor matrices is therefore determined solely by the original pixels; the padded positions contribute neither error nor gradient. In this way, the CP factors are fitted to the actual irregular region, and the zero-filled positions do not bias the decomposition result. A mask mechanism is introduced to map the low-rank approximation result $\mathcal{Y}_i$ back to the global tensor, which ensures that only the data in irregular regions participate in the final reconstruction, and accurately control the data participating in decomposition and reconstruction. In this way, a noise-free image is obtained through the global coordinate mapping reconstruction, which ensures that the aggregated image is consistent with the original overall consistency while preserving the spatial structure.

D. Iterative Optimization

1) *Objective Function:* After the above steps, the objective function of this method can be obtained as follows:

$$\begin{aligned} \min_{\{\mathcal{X}_i\}, \{\mathcal{S}_i\}} \quad & \sum_{i=1}^K (\|\mathcal{X}_i\|_{RDCP} + \lambda \|\mathcal{S}_i\|_1) \\ \text{s.t.} \quad & \mathcal{M}_i \odot \mathcal{Y}_i = \mathcal{M}_i \odot (\mathcal{X}_i + \mathcal{S}_i), \quad \forall i \end{aligned} \quad (15)$$

where $\mathcal{Y}_i \in \mathbb{R}^{h \times w \times B}$ represents the observed HSI data of the i_{th} superpixel region, $\mathcal{X}_i \in \mathbb{R}^{h \times w \times B}$ represents the low-rank part to be recovered, $\mathcal{S}_i \in \mathbb{R}^{h \times w \times B}$ represents the sparse part of the data (mixed noise), $\mathcal{M}_i$ represents the mask tensor, When $\mathcal{M}_i$ is 1, it indicates the original effective component; when $\mathcal{M}_i$ is 0, it means the supplementary part. $\odot$ represents element-wise multiplication. And λ represents the parameter of the sparse term l_1 norm regularization. K represents the number of splits determined in the superpixel adaptation step through homogeneous regions.

Due to the introduction of the mask mechanism to prevent the completed parts from interfering with the original image, only the positions where the mask is 1 should be used to update the factor matrix, ensuring that the low-rank decomposition is driven only by the original valid area. Where $\|\mathcal{X}_i\|_{RDCP}$ is defined as follows:

$$\|\mathcal{X}_i\|_{RDCP} = \min_{R_i} \left(\sum_{r=1}^{R_i} (\|a_r\|_2^2 + \|b_r\|_2^2 + \|c_r\|_2^2) \right)$$

$$\text{s.t. } \mathcal{X}_i \approx \sum_{r=1}^{R_i} \sigma_r \cdot a_r \circ b_r \circ c_r \quad (16)$$

Lagrange multiplier Λ_i and penalty parameter μ are introduced to construct the augmented Lagrange function as follows:

$$\begin{aligned} \mathcal{L}_\mu(\{\mathcal{X}_i\}, \{\mathcal{S}_i\}, \{\Lambda_i\}) = & \sum_{i=1}^K (\|\mathcal{X}_i\|_{RDCP} + \lambda \|\mathcal{M}_i \odot \mathcal{S}_i\|_1 \\ & + \langle \Lambda_i, \mathcal{M}_i \odot (\mathcal{Y}_i - \mathcal{X}_i - \mathcal{S}_i) \rangle \\ & + \frac{\mu}{2} \|\mathcal{M}_i \odot (\mathcal{Y}_i - \mathcal{X}_i - \mathcal{S}_i)\|_F^2) \end{aligned} \quad (17)$$

For the proposed model, the joint solving problem is decomposed into a number of independent subproblems. The Lagrange multiplier is introduced to solve the optimization sub-problems in a fixed order. When the convergence condition is reached, the iterative approximation of the optimal solution is obtained. Solving the overall model can be divided into the solution of three sub-problems, using the ADMM algorithm, which are the update problems of $\mathcal{X}_i$, $\mathcal{S}_i$ and Λ_i of each homogeneous region, and the specific optimization process is given as follows.

2) ADMM Optimization Process:

a) *Update low-rank terms $\mathcal{X}_i$:* Extracting all terms containing $\mathcal{X}_i^k$ in Eq (17) and introducing auxiliary variable Γ_i^k , the subproblem can be obtained as follows.

$$\mathcal{X}_i^{k+1} = \arg \min_{\mathcal{X}_i^k} \|\mathcal{X}_i^k\|_{RDCP} + \frac{\mu}{2} \|\mathcal{M}_i \odot (\mathcal{X}_i - \Gamma_i^k)\|_F^2 \quad (18)$$

$$\Gamma_i^k = \mathcal{Y}_i - \mathcal{X}_i^k + \frac{\Lambda_i^k}{\mu} \quad (19)$$

After Eq (16), Eq (18) and Eq (19), Eq (20) can be obtained:

$$\begin{aligned} \min_{\mathcal{A}_i, \mathcal{B}_i, \mathcal{C}_i} & \left(\sum_{r=1}^{R_i} (\|a_r\|_2^2 + \|b_r\|_2^2 + \|c_r\|_2^2) \right) \\ & + \frac{\mu}{2} \|\mathcal{M}_i \odot (\llbracket \mathcal{A}_i, \mathcal{B}_i, \mathcal{C}_i \rrbracket - \Gamma_i^k)\|_F^2 \end{aligned} \quad (20)$$

$$\text{s.t. } \mathcal{X}_i \approx \sum_{r=1}^{R_i} \sigma_r \cdot a_r \circ b_r \circ c_r \quad (21)$$

The classical alternating least squares method ALS is used to solve the CP factorization model, and each factor matrix can be solved along each mode by fixing the other factor matrices during optimization. R_i^k represents the optimal rank of the current region obtained by Eq (12) in the first iteration. The proposed method uses random initialization to obtain the factor matrices for the three modes $\mathcal{A}_i \in \mathbb{R}^{h \times R_i}$, $\mathcal{B}_i \in \mathbb{R}^{h \times R_i}$, $\mathcal{C}_i \in \mathbb{R}^{h \times R_i}$. a_r , b_r and c_r correspond to each column of the three factor matrices $\mathcal{A}_i$, $\mathcal{B}_i$ and $\mathcal{C}_i$. $\circ$ denotes the outer product of a number of rank-one vectors, from which the rank-one quantity is obtained, and summing the number R_i of rank-one quantities yields the desired low-rank part.

Specifically, fix $\mathcal{B}_i$ and $\mathcal{C}_i$ and update $\mathcal{A}_i$:

$$\mathcal{A}_i^{t+1} = \arg \min_{\mathcal{A}_i} \left(\sum_{r=1}^{R_i} \|a_r\|_2^2 + \frac{\mu}{2} \|\mathcal{A}_i (\mathcal{C}_i^t \odot \mathcal{B}_i^t)^\top - T_{i(1)}^t\|_F^2 \right) \quad (22)$$

Using the Khatri-Rao product, $T_{i(1)}^t$ represents the matrix obtained by the mode-1 expansion of the T_i^t tensor, and the

same expansion of mode-2 and mode-3 can be performed to obtain $T_{i(2)}^t$ and $T_{i(3)}^t$, where t represents the number of iterations of the ALS algorithm. For each homogeneous region $T_i^t \in \mathbb{R}^{h \times w \times B}$ has:

$$\begin{cases} T_{i(1)}^t \in \mathbb{R}^{h_i \times w_i B} \\ T_{i(2)}^t \in \mathbb{R}^{w_i \times h_i B} \\ T_{i(3)}^t \in \mathbb{R}^{B \times h_i w_i} \end{cases} \quad (23)$$

Then the analytical solution of Eq (22) can be computed as follows, and $\odot$ represents element-wise multiplication.

$$\mathcal{A}_i^{t+1} = T_{i(1)}^t (\mathcal{C}_i^t \odot \mathcal{B}_i^t) \left((\mathcal{C}_i^t)^\top \mathcal{C}_i^t * (\mathcal{B}_i^t)^\top \mathcal{B}_i^t + \frac{2}{\mu} \mathcal{I} \right)^{-1} \quad (24)$$

Similarly, update $\mathcal{B}$ and $\mathcal{C}$:

$$\mathcal{B}_i^{t+1} = T_{i(2)}^t (\mathcal{C}_i^t \odot \mathcal{A}_i^t) \left((\mathcal{C}_i^t)^\top \mathcal{C}_i^t * (\mathcal{A}_i^{t+1})^\top \mathcal{A}_i^{t+1} + \frac{2}{\mu} \mathcal{I} \right)^{-1} \quad (25)$$

$$\begin{aligned} \mathcal{C}_i^{t+1} = & T_{i(3)}^t (\mathcal{B}_i^{t+1} \odot \mathcal{A}_i^{t+1}) \left((\mathcal{B}_i^{t+1})^\top \mathcal{B}_i^{t+1} * (\mathcal{A}_i^{t+1})^\top \mathcal{A}_i^{t+1} \right. \\ & \left. + \frac{2}{\mu} \mathcal{I} \right)^{-1} \end{aligned} \quad (26)$$

After updating the factor matrices Eq (24), Eq (25), and Eq (26) of the three dimensions by the above steps, the updated low-rank tensor part can be obtained by reconstruction:

$$\mathcal{X}_i^{t+1} = \llbracket \mathcal{A}_i^{t+1}, \mathcal{B}_i^{t+1}, \mathcal{C}_i^{t+1} \rrbracket \quad (27)$$

The convergence condition of ALS is calculated based on the updated low-rank part Eq (27) as follows:

$$f^{t+1} = 1 - \frac{\|\mathcal{X}_i^{t+1} - \mathcal{X}_i^t\|_F}{\|\mathcal{X}_i^t\|_F}, \Delta f^{t+1} = |f^{t+1} - f^t| \quad (28)$$

b) *Update sparse terms $\mathcal{S}_i$:* Extracting all terms containing $\mathcal{S}_i$ from Equation Eq (17) yields:

$$\begin{aligned} \mathcal{S}_i^{k+1} = & \arg \min_{\mathcal{S}_i} \left(\lambda \|\mathcal{M}_i \odot \mathcal{S}_i\|_1 + \frac{\mu}{2} \|\mathcal{M}_i \odot (\mathcal{S}_i - \Delta_i^k)\|_F^2 \right) \\ \Delta_i^k = & \mathcal{Y}_i - \mathcal{X}_i^{k+1} + \frac{\Lambda_i^k}{\mu} \end{aligned} \quad (29)$$

Solving the problem by soft threshold operation can be obtained as follow:

$$\mathcal{S}_i^{k+1} = \mathcal{M}_i \odot \mathcal{S}_\eta(\Delta_i^k), \eta = \frac{\lambda}{\mu} \quad (30)$$

c) *Update the lagrange multiplier Λ_i :*

$$\Lambda_i^{k+1} = \Lambda_i^k + \mu \cdot \mathcal{M}_i \odot (\mathcal{Y}_i - \mathcal{X}_i^{k+1} - \mathcal{S}_i^{k+1}) \quad (31)$$

The overall algorithm flow of the proposed method is listed as Algorithm 1.

III. EXPERIMENTS

To ensure a fair comparison, all the comparison methods, including both the model-driven and the learning-based ones, are evaluated on the same machine, which is equipped with an Intel i9-12900K CPU and an NVIDIA RTX 4090 GPU; the model-driven methods are implemented in MATLAB R2024a, while the learning-based methods are executed within their official code environments. The parameter settings of the model-driven comparison methods are adjusted

Algorithm 1 The Proposed Method

Input: Observed HSI $\mathcal{Y}$, rank space $[R_{min}, R_{max}]$, sparsity regularization parameter λ

Initialization: $\alpha = 0.4$, $\beta = 0.035$, $\zeta = 1e-2$, $\omega = 0.5$, $\tau = 100$, $\rho = 4$, penalty factor of ADMM $\mu = 1e-10$, maximum number of iterations $\max_{iter} = 15$, $\xi = 1e-3$, maximum number of iterations of ALS $t_{max} = 50$

PCA: Adaptive segmentation of superpixels $ERS_k(Y)$ by (5)

Evaluating segmentation quality: Through (9) check the segmentation convergence condition $\|\text{SegQuality}_k - \text{SegQuality}_{k-1}\| < \zeta$

Approximation: Irregular tensor approximation and completion by (14)

While not converge **do**

- 1: ALS iterative update of CP decomposition
 - 1): Dynamically adjust the current regional optimal rank $\mathcal{R}_i^k$ by (12)
 - 2): Update $\Gamma_i^{k_t}$ by (18)
 - 3): Initialize $\mathcal{A}_i, \mathcal{B}_i, \mathcal{C}_i$
 - 4): Update $\mathcal{A}_i, \mathcal{B}_i, \mathcal{C}_i$ by (24), (25), (26)
- 2: Check the convergence condition $t \geq 2$ & $(t < t_{max} \|\Delta f^t\| < \xi)$, Update the number of ALS iterations $t = t + 1$
- 3: Update $\{X_i\}$ by (20), (21), (27)
- 4: Update $\{S_i\}$ by (29), (30)
- 5: Update $\{\Lambda_i\}$ by (31)
- 6: Check the convergence condition $k > \max_{iter}$ or $\|\mathcal{Y} - \mathcal{X} - \mathcal{S}\|_\infty < \xi$
- 7: Update the iteration count $k = k + 1$

end while

Output: $\mathcal{X} = \mathcal{X}_1 \oplus \mathcal{X}_2 \oplus \dots \oplus \mathcal{X}_i \oplus \dots \oplus \mathcal{X}_K$ is obtained by aggregation.

according to the values provided in their references. Specifically, besides the model-driven methods (LRTV, FallHyDe, SSCTV, 3DLogTNN, STCR, LRTDTV), we compare with four representative learning-based methods. For each learning-based method, we strictly follow the processing pipeline and configurations described in its original paper to guarantee its best performance: the supervised Transformer-based SERT [33] and SST [34] are evaluated with the official pre-trained weights released by their authors, which are trained on the ICVL dataset (31 spectral bands); the two recent unsupervised methods, HIR-Diff [35] and S2DIP [36], which require no external training data, are run with the default hyperparameters recommended in their papers—HIR-Diff restores the image using a pre-trained diffusion prior, while S2DIP optimizes a double-subspace deep prior directly on each test image.

1) *Simulated Experiment:* In this section, two measures of spatial recovery quality and two measures of spectral recovery quality are used to quantitatively and comprehensively evaluate the performance. MPSNR: Mean Peak Signal-to-Noise Ratio, MSSIM: Mean Structural Similarity, ERGAS: Error Relative Global Accuracy Synthesis, SAM: Spectral Angle Mapping.

A better denoising effect is indicated by higher MPSNR and MSSIM values and lower ERGAS and SAM values. The noise-free HSI data of WDC and WHU-HongHu after clipping are selected as the experimental objects with simulated noise added, and the data size is $200 \times 200 \times 270$ and $200 \times 200 \times 191$, respectively. The noise scenarios are described as follows:

Case 1: All bands are contaminated by additive Gaussian noise with zero mean. The standard deviation of Gaussian noise is randomly distributed between $[0.1, 0.2]$.

Case 2: Gaussian noise is added in the same way as Case 1 with a standard deviation between $[0.1, 0.2]$. In addition, impulsive noise with standard deviation randomly distributed between $[0.1, 0.2]$ was added to all bands.

Case 3: On the basis of Case 2, 40% of the bands are randomly selected to add dead lines in the range of $[1-15]$ respectively.

Case 4: On the basis of Case 2, 40% of the bands are randomly selected to add stripe noise in the range of $[10-20]$ respectively.

Case 5: Based on Case 2, dead line and stripe noise with the same intensity and number of lines as Case 3 and Case 4 are randomly added.

In addition to providing intuitive false color visualization comparison results, this section also presents the comparison results of four quantitative metrics in Table I and Table II. The best results in each case are highlighted in **bold format**, while the second-best metrics are in underlined format. The detailed experimental results and analysis are described as follows.

In Table I, the proposed method consistently achieves the best MPSNR, MSSIM and ERGAS across all five noise cases, demonstrating a clear advantage in spatial-structural restoration and global error control. For the spectral metric SAM, the proposed method ranks first in the complex-noise cases (Case 3-5); in the light-noise Case 1 both the unsupervised diffusion-based HIR-Diff and the deep-prior-based S2DIP achieve a lower SAM than the proposed method, and HIR-Diff is also lower in Case 2, owing to their explicit spectral-subspace modeling. Across all four indicators, however, the proposed method remains the most balanced, leading on at least three of the four metrics in every case. In the complex noise scenes such as Case 4-5, the proposed method maintains stable performance, whereas other methods, such as 3DLogTNN [37] and STCR [38], fluctuate greatly.

As shown in Table II, the STCR method, which considers the low-rank property of the full mode, performs best in removing the structurally strong noise of Case 1-2. When the noise scene becomes complex (Case 3-5), STCR does not account for the mixed-noise characteristics, so its performance degrades on dense dead lines and stripe noise and its robustness is weak; in these complex cases the proposed method attains the best MPSNR and MSSIM, confirming its superior spatial-structural restoration. For the spectral metrics on WDC, the unsupervised HIR-Diff [35] achieves slightly better SAM and ERGAS in the complex-noise cases, as its low-rank spectral subspace coupled with a pre-trained diffusion prior favors spectral-angle preservation; however, the gap is small (e.g., SAM 6.264 vs. 6.661 in Case 5), and the proposed method leads consistently on the spatial-fidelity indicators MPSNR

TABLE I
COMPARISON OF QUANTITATIVE INDEXES OF THE SIMULATED WHU-HONGHU HSI DATA

Case	Indexes	Noisy	LRTV	FallHyDe	SSCTV	LogTNN	STCR	LRTDTV	SERT	SST	HIR-Diff	S2DIP	Proposed
Case 1	MPSNR	33.72	34.77	35.30	35.58	31.99	<u>36.56</u>	30.69	32.67	32.38	32.97	32.73	38.81
	MSSIM	0.865	0.952	0.960	0.966	<u>0.983</u>	<u>0.976</u>	0.932	0.906	0.903	0.942	0.924	0.989
	ERGAS	156.9	109.1	140.1	140.1	121.2	135.6	55.50	101.4	104.4	68.85	<u>53.29</u>	44.17
	SAM	19.58	2.837	2.427	1.989	1.952	2.106	1.656	2.421	2.125	<u>1.500</u>	1.417	1.933
Case 2	MPSNR	17.90	34.77	31.59	<u>37.94</u>	36.80	37.56	37.70	33.23	32.56	<u>36.24</u>	32.61	39.16
	MSSIM	0.392	0.963	0.895	0.947	<u>0.984</u>	0.967	0.947	0.928	0.927	0.957	0.896	0.985
	ERGAS	488.9	70.58	102.8	49.17	78.61	60.38	<u>49.01</u>	82.12	86.60	51.29	77.80	35.08
	SAM	27.61	4.829	5.864	3.944	2.954	2.989	3.128	4.521	4.236	2.362	3.086	2.797
Case 3	MPSNR	16.92	34.28	28.51	<u>36.27</u>	27.08	35.60	34.36	32.24	31.66	36.10	30.44	39.17
	MSSIM	0.356	<u>0.958</u>	0.831	0.930	0.844	0.950	0.943	0.923	0.916	0.956	0.829	0.980
	ERGAS	507.7	<u>73.96</u>	125.56	60.64	166.20	57.55	71.04	95.04	104.5	51.78	98.56	35.83
	SAM	29.78	3.901	6.261	4.136	6.411	2.865	4.471	3.596	3.432	<u>2.342</u>	3.656	1.997
Case 4	MPSNR	17.08	30.02	27.19	34.15	<u>36.47</u>	35.60	20.31	29.57	26.95	36.11	32.36	38.90
	MSSIM	0.371	0.883	0.831	0.904	<u>0.924</u>	0.915	0.941	0.839	0.816	<u>0.955</u>	0.891	0.984
	ERGAS	544.4	241.6	261.0	283.8	240.0	283.7	59.16	136.0	189.7	<u>51.52</u>	79.45	28.92
	SAM	29.96	3.829	4.769	3.711	6.517	2.787	4.567	4.237	3.981	<u>2.407</u>	3.117	1.682
Case 5	MPSNR	16.08	27.64	23.64	31.50	24.61	30.42	35.67	28.70	26.20	<u>36.10</u>	30.31	38.85
	MSSIM	0.329	0.835	0.703	0.870	0.772	0.858	0.936	0.831	0.806	<u>0.954</u>	0.831	0.977
	ERGAS	576.9	324.1	362.7	321.7	331.9	324.5	69.85	142.1	205.7	<u>51.85</u>	99.61	39.72
	SAM	32.97	3.894	7.069	4.489	7.608	3.793	5.181	5.214	5.653	<u>2.385</u>	3.690	1.982

TABLE II
COMPARISON OF QUANTITATIVE INDEXES ON THE SIMULATED WDC HSI DATA

Case	Indexes	Noisy	LRTV	FallHyDe	SSCTV	LogTNN	STCR	LRTDTV	SERT	SST	HIR-Diff	S2DIP	Proposed
Case 1	MPSNR	28.48	33.13	34.88	36.91	31.86	38.46	33.87	31.06	31.54	36.37	33.84	<u>37.12</u>
	MSSIM	0.716	0.940	0.970	0.950	0.954	0.982	0.938	0.918	0.922	0.967	0.947	<u>0.973</u>
	ERGAS	219.6	103.0	97.42	61.15	124.8	68.26	77.26	127.6	132.7	59.36	77.41	67.53
	SAM	17.06	7.075	6.967	5.594	5.406	5.913	4.522	6.562	6.148	<u>4.210</u>	4.115	4.686
Case 2	MPSNR	17.68	33.18	29.50	36.95	33.09	37.12	33.53	30.65	30.94	34.86	30.71	<u>36.98</u>
	MSSIM	0.421	0.940	0.835	0.944	0.962	0.964	0.935	0.909	0.914	0.956	0.884	0.970
	ERGAS	601.4	103.8	158.2	<u>66.07</u>	108.1	55.49	80.50	128.0	127.1	70.01	115.73	67.78
	SAM	32.30	7.179	8.108	5.459	5.949	3.972	5.689	7.841	7.259	4.370	6.380	<u>4.293</u>
Case 3	MPSNR	16.88	32.95	25.28	<u>35.58</u>	28.87	33.69	33.18	30.17	30.55	34.81	29.02	36.01
	MSSIM	0.393	0.938	0.695	0.941	0.878	0.935	0.932	0.903	0.908	<u>0.954</u>	0.832	0.965
	ERGAS	617.2	105.6	239.8	69.93	151.4	81.50	84.15	135.0	130.6	<u>70.11</u>	135.98	73.44
	SAM	34.46	7.128	11.59	6.119	6.792	<u>5.401</u>	6.817	8.239	8.568	4.237	6.774	6.071
Case 4	MPSNR	16.25	33.00	26.05	<u>35.72</u>	30.77	29.10	32.54	23.69	23.03	34.64	30.44	36.64
	MSSIM	0.387	0.939	0.716	0.937	0.847	0.817	0.914	0.723	0.703	<u>0.953</u>	0.875	0.968
	ERGAS	725.6	103.4	315.5	72.48	437.4	514.1	142.9	326.8	349.4	<u>71.66</u>	116.87	70.11
	SAM	37.61	7.093	11.65	6.137	8.952	7.254	9.925	10.22	11.04	4.884	6.485	6.106
Case 5	MPSNR	15.86	32.79	23.18	<u>34.88</u>	24.94	27.69	32.56	23.58	22.57	34.78	28.89	35.21
	MSSIM	0.359	0.937	0.628	0.933	0.772	0.795	0.919	0.718	0.690	<u>0.954</u>	0.827	0.958
	ERGAS	727.2	106.2	502.6	81.16	447.4	566.7	122.4	321.3	367.3	72.35	137.99	80.27
	SAM	39.01	7.211	12.38	6.854	9.458	7.705	10.44	12.30	12.83	6.264	6.913	<u>6.661</u>

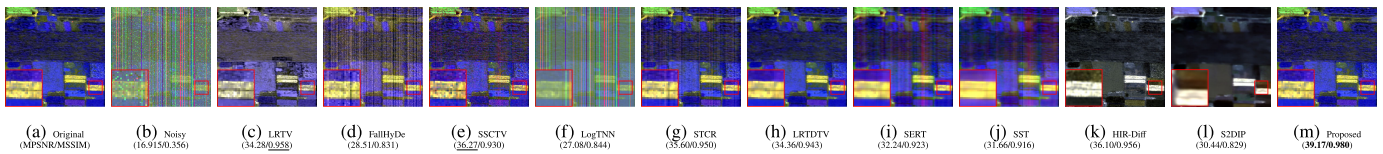


Fig. 2. Pseudo-color denoising results for (103, 48, 9) bands of WHU-HongHu HSI data in Case 3.

and MSSIM in the complex-noise cases, while requiring no pre-trained generative model or per-image network training. To further compare the reconstruction quality, this section shows the comparison of visual results for Case 3 in Table I and Case 4 in Table II. As shown in Fig. 2 and Fig. 3, FallHyDe [39], STCR, SERT [33], and SST [34] all suffer

from severe noise residual, in addition to which SERT and SST lose a great deal of texture detail information. Although LRTV [15] and SSCTV [40] perform well in noise removal, they suffer from serious spectral distortion problems. Both the proposed method and LRTDTV [15] achieve considerable results in removing stripe and dead line, but the proposed

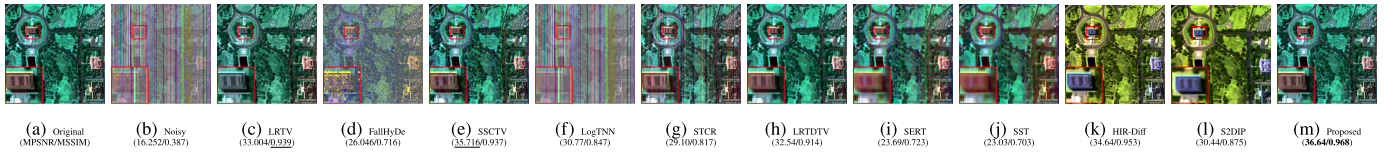


Fig. 3. Pseudo-color denoising results of (30, 111, 164) bands of WDC HSI data in Case 4.

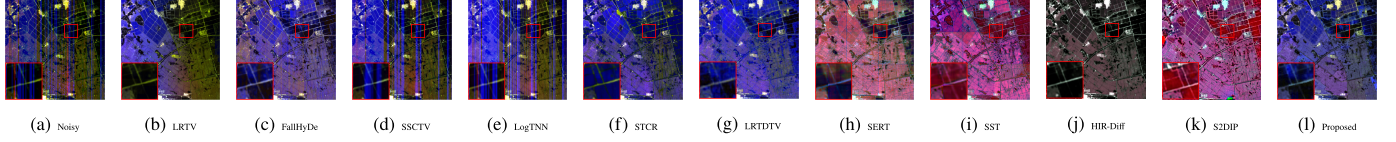


Fig. 4. Pseudo-color denoising results of the (27, 60, 49) bands of ZY-1 02D HSI data.

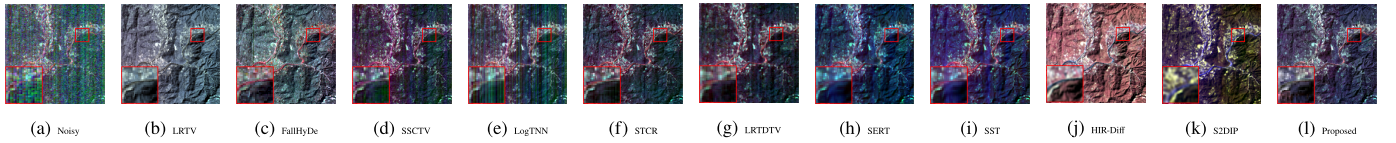


Fig. 5. Pseudo-color denoising result of band (1, 132, 166) of EO-1 HSI data.

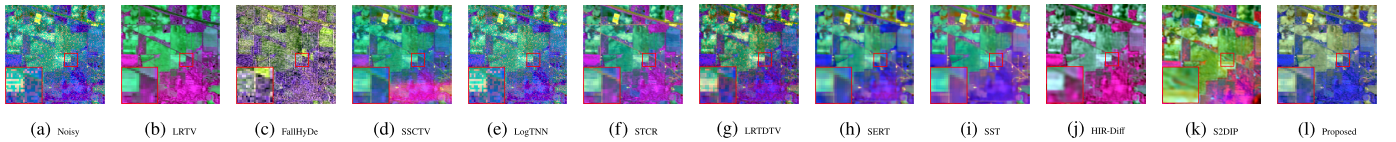


Fig. 6. Pseudo-color denoising results for (1, 17, 65) bands of Indian Pines data.

method is better than LRTDTV in spectral preservation. It can be shown that the proposed method not only has excellent image structure preservation ability, but also has strong global error control ability.

A. Real Experiments

In order to further evaluate the denoising effect of the proposed method on mixed noise HSI in real scenes, this section conducts experiments on four real HSI data: ZY-1 02D ($400 \times 400 \times 90$), EO-1 ($201 \times 200 \times 166$), Indian Pines ($145 \times 145 \times 200$) and GF-5 ($201 \times 201 \times 330$) HSI satellite data. The selected experimental data are contaminated by different kinds of mixed noise.

Fig. 4 gives the denoising visualization result of (27, 60, 49) bands of ZY-1 02D data. These three selected bands are very representative: they are severely damaged by dense dead lines, stripe noise pollution. The noise removal of this data is very challenging. It can be observed that SSCTV and 3DLogTNN still have a lot of dead lines remaining, while FallHyDe has a lot of stripe noise remaining and a vertical line texture appears. LRTDTV and STCR can remove part of the dead lines well, but the spectral distortion is caused to a certain extent. LRTV has a relatively good effect on removing dead lines, but it loses a lot of texture detail information. The deep learning-based methods SERT and SST produce severe spectral distortion with severe information loss. It can be seen in Fig. 4 that compared with other methods, the proposed method shows obvious advantages in the recovery of the mixed noise of dead lines and stripes.

Then, the (1, 132, 166) bands of EO-1 data are selected to synthesize the pseudo-color image. It is contaminated by a large amount of mixed noise, and the striping noise is the most serious. From the Fig. 5, except for the proposed method, the results of the comparison methods show the phenomenon of different degrees of spectral distortion and noise residual.

Next, the pseudo-color image synthesized by (1, 17, 65) and the grayscale image of the 327-th band are also selected for Indian Pines and GF-5, respectively. As shown in Fig. 6, in the results of Indian Pines, FallHyDe and 3DLogTNN all have different degrees of noise residual problems. There are serious spectral distortions in LRTV, STCR and SSCTV. SERT and SST, have a good effect on removing noise. However, there is the phenomenon of detail texture over-smoothing. The proposed method can effectively remove the noise and recover clearer texture details at the same time, showing obvious advantages in spectral reconstruction. Spectral curves of the point (50, 75) of Indian Pines are in Fig. 7. Compared with the SOTA methods, the proposed method can achieve reliable spectral results while maintaining the spectrum.

In the reconstruction results of GF-5, as shown in Fig. 8, SERT and SST also have the problem of detail over-smoothing. FallHyDe, LRTV, 3DLogTNN and STCR show different degrees of spectral distortion. The unsupervised HIR-Diff and S2DIP suppress most of the noise but still leave some residual artifacts, while LRTDTV, SSCTV and the proposed method achieve good denoising results.

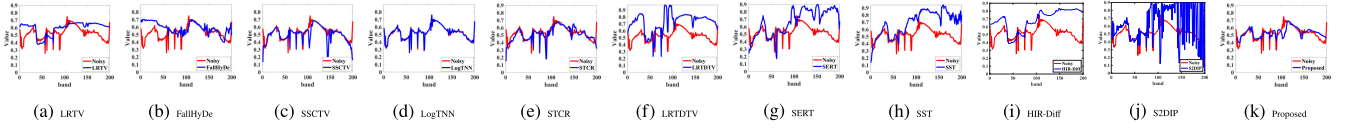


Fig. 7. Spectral curve of the point (50, 75) of Indian Pines before and after applying different denoising methods.

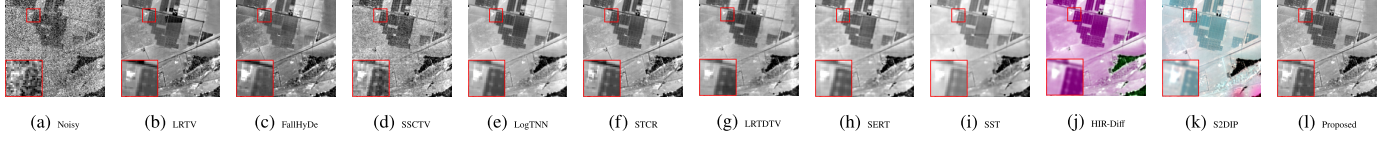
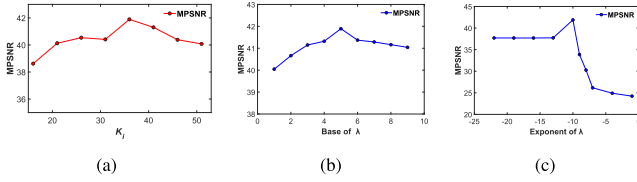
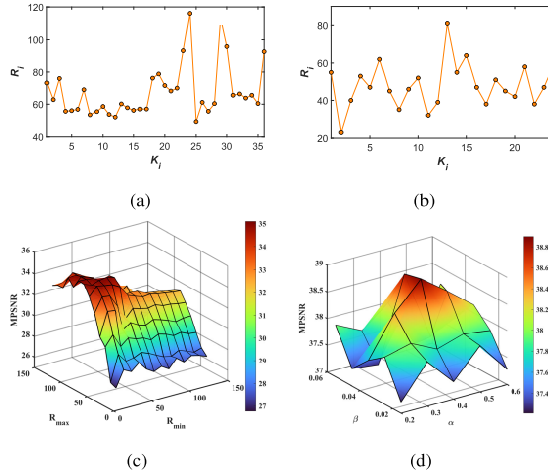


Fig. 8. Pseudo-color denoising results for (1, 17, 65) bands of GF-5 data.

Fig. 9. Effectiveness analysis of parameter adaptation of WHU-HongHu Case 4. (a) Effectiveness of K_i . (b) Base analysis of λ . (c) Exponent analysis of λ .Fig. 10. Effectiveness analysis of parameter adaptation. (a) Interval analysis of regional R_i of WHU-HongHu Case 4. (b) Interval analysis of regional R_i of WDC Case 5. (c) WDC Case 5 Rank Space Size Sensitivity Analysis. (d) Parameters sensitivity analysis in SegQ of WDC Case 5.

IV. DISCUSSION

A. Effectiveness Analysis of Parameter Adaptation

The proposed method has two adaptive parameters that will affect the denoising results, K and $[R_{min}, R_{max}]$, respectively. Where K is the number of segmentations determined by the proposed adaptive strategy. $[R_{min}, R_{max}]$ is the mapping space used by the framework of dynamic adjustment of region rank number, and to verify that the values obtained by adaptation are the optimal parameters, the two parameters are cross-validated on the WHU-HongHu Case 4. Different segmentation numbers K were set in Fig. 9 (a), when K is the number obtained through adaptive calculation, a

TABLE III
ANALYSIS OF THE UPPER BOUND ON R_{max}

Index	$R_{max} = 70$	$R_{max} = 80$
MPSNR	41.89	41.45
MSSIM	0.984	0.983

reliable MPSNR was achieved. Moreover, this section also carries out statistical analysis of R_i of each homogeneous region obtained on data WHU-HongHu Case 4 and WDC Case 5. The horizontal axis in Fig. 10 (a) and (b) represent the sequence number of homogeneous region, and the vertical axis represents the mean value of region rank in each iteration process. The average rank number of each region falls in the interval $[40, 70]$ at most, and some ranks of WHU-HongHu Case4 fall within 80, so we set the spatial upper bound of rank to 70 and 80 respectively for comparison, and the lower bound of rank is set to 40, and the comparison results are shown in Table III. The upper bound of rank is 70, the quantitative index obtained is better. Setting the rank range between 40 and 70 yields reliable results

B. Parameter Sensitivity Analysis

In addition, there is a sparsity regularization parameter λ which has a great impact on the denoising effect. By imposing the sparsity constraint, the algorithm will tend to retain the sparse noise-free part while suppressing the small and scattered noise, so as to realize the separation of the noise-free part and the noise in the transform domain. The sparsity constraint can adaptively deal with different regions, sparse flat regions are more smooth, while complex regions retain more details. To verify the effectiveness of the selected λ during the experiment, we also verified λ in WHU-HongHu Case 4. Since λ uses the representation form of science and technology method, we cross-validated the Base and Exponent in science counting method, and the results are shown in Fig. 9 (b) and Fig. 9 (c), respectively. The horizontal axis represents the cardinality of the parameter, and the vertical axis represents the quantitative index MPSNR results. It can be seen from the figure that the best experimental results can be obtained when the cardinality and exponent of λ are $[5, -10]$ respectively.

TABLE IV
DENOISING TIME AND CLASSIFICATION ACCURACY INDEXES FOR DOWNSTREAM TASKS

Dataset	Index	LRTV	FallHyDe	SSCTV	LogTNN	STCR	LRTDTV	SERT	SST	HIR-Diff	S2DIP	Proposed
Indian Pines	OA	0.884	0.579	<u>0.920</u>	0.795	0.889	0.855	0.826	0.835	0.892	0.819	0.927
	Kappa	0.868	0.517	<u>0.907</u>	0.767	0.873	0.834	0.817	0.813	0.822	0.815	0.916
	Time/s	83	0.983	120	67	10	35	12	11	11	60	129
Pavia University	OA	<u>0.972</u>	0.937	0.963	0.933	0.967	0.936	0.969	<u>0.972</u>	0.951	0.917	0.978
	Kappa	0.969	0.907	0.951	0.911	0.956	0.912	0.967	<u>0.970</u>	0.957	0.913	0.971
	Time/s	275	1.972	1757	84	59	159	11	10	37	684	714

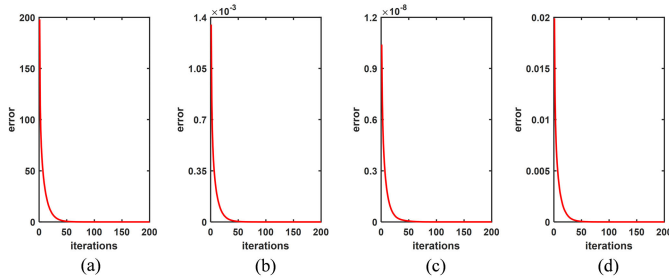


Fig. 11. Convergence curves of proposed model on four datasets. (a) ZY-1 02D. (b) EO1. (c) Indian Pines. (d) GF-5.

C. Convergence Analysis

The convergence curves of the four real experiments are respectively represented in Fig. 11. The algorithm demonstrated stable convergence characteristics within a limited number of iterations, with the residuals continuously decreasing and stabilizing. Therefore, the effectiveness of the proposed algorithm has been strongly demonstrated.

D. Algorithm Complexity and Downstream Tasks

Since the adaptive capabilities of superpixel segmentation and low-rank decomposition need to be achieved through iterations, the time complexity of this method is $\mathcal{O}(N \log N + Nr^2)$. Specifically, N denotes the total number of spatial pixels in the hyperspectral image. $\mathcal{O}(N \log N)$ term mainly arises from the greedy algorithm used in the entropy rate superpixel segmentation. The variable r represents the dynamically adjusted regional rank. $\mathcal{O}(Nr^2)$ term corresponds to the computational cost of updating the factor matrices along the three modes during the ALS optimization. The classification indicators and the time required for denoising of the real experiment Indian Pines and Pavia University are in Table IV. The OA and kappa values of the proposed method are at the leading position, which indicates that it is beneficial for downstream tasks. Although the running time of the proposed method was slightly longer, the significant improvement in the performance of downstream tasks offset the time cost, which demonstrates the trade-off between efficiency and effectiveness.

The number of superpixels segmented and the running time calculated by the experiment for each experiment data are recorded in Table V, Table VI and Table VII. From the table, it can be seen that the computing time of this method is not entirely determined by the partitioning number. To a certain extent, it also depends on the size of the data itself. As can be

TABLE V
RECORDS OF K AND RUNNING TIME FOR REAL DATA

Value	ZY-1 02D	EO-1	Indian Pines	GF-5
K	36	46	28	64
Time/s	435	289	154	343

TABLE VI
RECORDS OF K AND RUNNING TIME FOR WHU-HONGHU

Value	Case 1	Case 2	Case 3	Case 4	Case 5
K	36	36	36	36	36
Time/s	371	378	278	247	255

TABLE VII
RECORDS OF K AND RUNNING TIME FOR WDC

Value	Case 1	Case 2	Case 3	Case 4	Case 5
K	26	26	25	26	24
Time/s	190	187	295	319	325

seen from the computing time of GF-5, when the number of bands is large, the computing time will also increase. Within an acceptable range, the feasibility of the proposed method can be concluded.

V. CONCLUSION

In this paper, we propose an HSI denoising method based on superpixel adaptive segmentation and dynamic adjustment of region optimal rank. The proposed method fully exploits the global spatial-spectral self-similarity of the image, and avoids the influence of empirical parameters on the clustering accuracy. Although exact CP rank determination is NP-hard, the bounded regression used here provides a practical sub-optimal estimate. Experiments on six sensors with different spectral ranges and noise characteristics confirm that the method generalizes across datasets rather than being tied to a single scenario. The proposed method still has certain limitations. In cases where the noise is overly complex and intense, the quality and quantity of the segmentation may be affected, resulting in prolonged computation time and increased complexity. In future work, we plan to adopt more advanced noise-aware technologies to adjust the sorting of regions, and study optimization algorithms to accelerate the computation speed.

REFERENCES

- [1] X. Lu, W. Zhang, and X. Li, "A hybrid sparsity and distance-based discrimination detector for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1704–1717, Mar. 2018.
- [2] Y. Yuan, M. Fu, and X. Lu, "Low-rank representation for 3D hyperspectral images analysis from map perspective," *Signal Process.*, vol. 112, pp. 27–33, Jul. 2015.
- [3] Q. Zhang, J. Zhu, Y. Dong, E. Zhao, M. Song, and Q. Yuan, "10-minute forest early wildfire detection: Fusing multi-type and multi-source information via recursive transformer," *Neurocomputing*, vol. 616, Feb. 2025, Art. no. 128963.
- [4] Q. Zhang, J. Zhu, Y. Huang, Q. Yuan, and L. Zhang, "Beyond being wise after the event: Combining spatial, temporal and spectral information for Himawari-8 early-stage wildfire detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 124, Nov. 2023, Art. no. 103506.
- [5] Y. Chen, X. Cao, Q. Zhao, D. Meng, and Z. Xu, "Denoising hyperspectral image with non-i.i.d. noise structure," *IEEE Trans. Cybern.*, vol. 48, no. 3, pp. 1054–1066, Mar. 2018.
- [6] Z. Lai, C. Yan, and Y. Fu, "Hybrid spectral denoising transformer with guided attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 13019–13029.
- [7] G. Meng et al., "Sp3ctralMamba: Physics-driven joint state space model for hyperspectral image reconstruction," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2025, vol. 39, no. 6, pp. 6108–6116.
- [8] Q. Zhang, Y. Zheng, Q. Yuan, M. Song, H. Yu, and Y. Xiao, "Hyperspectral image denoising: From model-driven, data-driven, to model-data-driven," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 13143–13163, Oct. 2024.
- [9] Q. Zhang, Q. Yuan, J. Li, Z. Li, H. Shen, and L. Zhang, "Thick cloud and cloud shadow removal in multitemporal imagery using progressively spatio-temporal patch group deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 148–160, Apr. 2020.
- [10] M. Elad and M. Aharon, "Image denoising via sparse and redundant representations over learned dictionaries," *IEEE Trans. Image Process.*, vol. 15, no. 12, pp. 3736–3745, Dec. 2006.
- [11] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.
- [12] X. Cao, Q. Zhao, D. Meng, Y. Chen, and Z. Xu, "Robust low-rank matrix factorization under general mixture noise distributions," *IEEE Trans. Image Process.*, vol. 25, no. 10, pp. 4677–4690, Oct. 2016.
- [13] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.
- [14] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral-spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.
- [15] Y. Wang, J. Peng, Q. Zhao, Y. Leung, X.-L. Zhao, and D. Meng, "Hyperspectral image restoration via total variation regularized low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1227–1243, Apr. 2018.
- [16] Q. Zhang, Y. Dong, Q. Yuan, M. Song, and H. Yu, "Combined deep priors with low-rank tensor factorization for hyperspectral image restoration," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
- [17] Q. Zhang et al., "Three-dimension spatial-spectral attention transformer for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531213.
- [18] X. Liu, S. Bourennane, and C. Fossati, "Denoising of hyperspectral images using the PARAFAC model and statistical performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3717–3724, Oct. 2012.
- [19] Q. Zhang, Q. Yuan, M. Song, H. Yu, and L. Zhang, "Cooperated spectral low-rankness prior and deep spatial prior for HSI unsupervised denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 6356–6368, 2022.
- [20] Q. Zhang, Y. Zheng, Y. Dong, C. Yu, and Q. Yuan, "Hyperspectral image mixed noised removal via jointly spatial and spectral difference constraint with low-rank tensor factorization," *Eng. Appl. Artif. Intell.*, vol. 149, Jun. 2025, Art. no. 110508.
- [21] W. He, Q. Yao, C. Li, N. Yokoya, and Q. Zhao, "Non-local meets global: An integrated paradigm for hyperspectral denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 6861–6870.
- [22] Y. Peng, D. Meng, Z. Xu, C. Gao, Y. Yang, and B. Zhang, "Decomposable nonlocal tensor dictionary learning for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2949–2956.
- [23] Q. Xie et al., "Multispectral images denoising by intrinsic tensor sparsity regularization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 1692–1700.
- [24] J. Xue, Y. Zhao, W. Liao, and J. C.-W. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.
- [25] S. Zhang, S. Li, W. Fu, and L. Fang, "Multiscale superpixel-based sparse representation for hyperspectral image classification," *Remote Sens.*, vol. 9, no. 2, p. 139, Feb. 2017.
- [26] J. Jiang, J. Ma, C. Chen, Z. Wang, Z. Cai, and L. Wang, "SuperPCA: A superpixelwise PCA approach for unsupervised feature extraction of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4581–4593, Aug. 2018.
- [27] Y. Peng, Y. Zhang, W. Kong, F. Nie, B.-L. Lu, and A. Cichocki, "S3LRR: A unified model for joint discriminative subspace identification and semisupervised EEG emotion recognition," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.
- [28] X. Mei, Y. Ma, C. Li, F. Fan, J. Huang, and J. Ma, "Robust GBM hyperspectral image unmixing with superpixel segmentation based low rank and sparse representation," *Neurocomputing*, vol. 275, pp. 2783–2797, Jan. 2018.
- [29] B. Han, Y. Jia, H. Liu, and J. Hou, "Irregular tensor low-rank representation for hyperspectral image representation," *IEEE Trans. Image Process.*, vol. 34, pp. 3239–3252, 2025.
- [30] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 2097–2104.
- [31] X. Liu, M. Tanaka, and M. Okutomi, "Single-image noise level estimation for blind denoising," *IEEE Trans. Image Process.*, vol. 22, no. 12, pp. 5226–5237, Dec. 2013.
- [32] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862–2869.
- [33] M. Li, J. Liu, Y. Fu, Y. Zhang, and D. Dou, "Spectral enhanced rectangle transformer for hyperspectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Apr. 2023, pp. 5805–5814.
- [34] M. Li, Y. Fu, and Y. Zhang, "Spatial-spectral transformer for hyperspectral image denoising," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 1368–1376.
- [35] L. Pang, X. Rui, L. Cui, H. Wang, D. Meng, and X. Cao, "HIR-Diff: Unsupervised hyperspectral image restoration via improved diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3005–3014.
- [36] K. Shi, J. Peng, J. Gao, Y. Luo, and S. Xu, "Hyperspectral image denoising via double subspace deep prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531015.
- [37] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, T.-X. Jiang, T.-H. Ma, and T.-Y. Ji, "Mixed noise removal in hyperspectral image via low-fibered-rank regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 734–749, Jan. 2020.
- [38] L. Sun, C. He, Y. Zheng, Z. Wu, and B. Jeon, "Tensor cascaded-rank minimization in subspace: A unified regime for hyperspectral image low-level vision," *IEEE Trans. Image Process.*, vol. 32, pp. 100–115, 2023.
- [39] Y. Chen, J. Zeng, W. He, X.-L. Zhao, T.-X. Jiang, and Q. Huang, "Fast large-scale hyperspectral image denoising via noniterative low-rank subspace representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530714.
- [40] J. Gao, H. Wang, and J. Peng, "Hyperspectral image restoration via RPCA model based on spectral-spatial correlated total variation regularizer," *Neurocomputing*, vol. 619, Feb. 2025, Art. no. 128885.